

Topic 4A: The elements of Groups 1 and 2
1. understand reasons for the trend in ionisation energy down Group 2
2. understand reasons for the trend in reactivity of the Group 2 elements down the group
3. know the reactions of the elements Mg to Ba in Group 2 with oxygen, chlorine and water
4. know the reactions of the oxides of Group 2 elements with water and dilute acid, and their hydroxides with dilute acid
5. know the trends in solubility of the hydroxides and sulfates of Group 2 elements
6. understand reasons for the trends in thermal stability of the nitrates and the carbonates of the elements in Groups 1 and 2 in terms of the size and charge of the cations involved
7. understand the formation of characteristic flame colours by Group 1 and 2 compounds in terms of electron transitions <i>Students will be expected to know the flame colours for Groups 1 and 2 compounds.</i>
8. understand experimental procedures to show: i patterns in thermal decomposition of Group 1 and 2 nitrates and carbonates ii flame colours in compounds of Group 1 and 2 elements